As an analogy, consider the following context where humans also occasionally fabricate plausible-sounding information. When uncertain, students may guess on multiple-choice exams and even bluff on written exams, submitting plausible answers in which they have little confidence. Language models are evaluated by similar tests. In both settings, guessing when unsure maximizes the expected score under a binary 0-1 scheme that awards 1 point for a correct answer and none for blanks or IDKs. Bluffs are often overconfident and specific, such as "September 30" rather than "Sometime in autumn" for a question about a date. Many language-model benchmarks mirror standardized human exams, using binary metrics such as accuracy or pass-rate. Optimizing models for these benchmarks may therefore foster hallucinations. Humans learn the value of expressing uncertainty outside of school, in the school of hard knocks. On the other hand, language models are primarily evaluated using exams that penalize uncertainty. Therefore, they are always in "test-taking" mode. Put simply, most evaluations are not aligned.

We are not the first to realize that binary grading does not measure hallucination. However, prior work on hallucination evaluation has generally sought after the elusive "perfect hallucination eval." In Section 4, we argue that this is insufficient. We observe that existing primary evaluations overwhelmingly penalize uncertainty, and thus the root problem is the *abundance of evaluations that are not aligned*. Suppose Model A is an aligned model that correctly signals uncertainty and never hallucinates. Let Model B be similar to Model A except that it never indicates uncertainty and always "guesses" when unsure. Model B will outperform A under 0-1 scoring, the basis of most current benchmarks. This creates an "epidemic" of penalizing uncertainty and abstention, which we argue that a small fraction of hallucination evaluations won't suffice. The numerous primary evaluations must be adjusted to stop penalizing abstentions when uncertain.

Contributions. We identify the main statistical drivers of hallucinations, from their pretraining origins to their post-training persistence. A novel connection between supervised and unsupervised learning demystifies their origin, even when training data contain IDK. The persistence of hallucinations, despite extensive work on the problem, is explained by the recognition that hallucination-like guessing is rewarded by most primary evaluations. We discuss statistically rigorous modifications to existing evaluations that pave the way to effective mitigation.

2 Related work

To the best of our knowledge, the reduction from supervised learning (binary classification) to unsupervised learning (density estimation or self-supervised learning) presented in this work is novel. The general method of reduction between learning problems, however, is a well-established technique for demonstrating that one problem is at least as hard as another (see, e.g., Beygelzimer et al., 2016).

A number of surveys and studies have explored the underlying causes of hallucination in language models. Sun et al. (2025) cite factors such as model overconfidence Yin et al. (2023), decoding randomness Lee et al. (2022), snowballing effects Zhang et al. (2023), long-tailed training samples Sun et al. (2023), misleading alignment training Wei et al. (2023), spurious correlations Li et al. (2022), exposure bias Bengio et al. (2015), the reversal curse Berglund et al. (2024), and context hijacking Jeong (2024). Analogous sources of error have long been studied in broader machine learning and statistical settings (Russell and Norvig, 2020).

The most closely related theoretical work is by Kalai and Vempala (2024), which we show is

a special case of our reduction. They connected the Good-Turing missing mass estimates (Good, 1953) to hallucinations, which inspired Theorem 3. However, that work does not address uncertainty expressions (e.g., IDK), connections to supervised learning, post-training modifications, and their model did not include prompts. Hanneke et al. (2018) analyze an interactive learning algorithm that queries a validity oracle (e.g., a human) to agnostically train a language model that minimizes hallucinations. Their method is statistically efficient, requiring a reasonable amount of data, but not computationally efficient. Other recent theoretical studies (Kalavasis et al., 2025; Kleinberg and Mullainathan, 2024) formalize an inherent trade-off between *consistency* (avoiding invalid outputs) and *breadth* (generating diverse, linguistically rich content). These works demonstrate that for broad classes of languages, any model that generalizes beyond its training data will either hallucinate invalid outputs or suffer mode collapse, failing to produce the full range of valid responses.

Several post-training techniques—such as reinforcement learning from human feedback (RLHF) (Ouyang et al., 2022), reinforcement learning from AI feedback (RLAIF) (Bai et al., 2022), and direct preference optimization (DPO) (Rafailov et al., 2023)—have been shown to reduce hallucinations, including conspiracy theories and common misconceptions. Gekhman et al. (2024) show that simple fine-tuning on novel information can initially decrease hallucination rates, only for them to later increase. Further, it has been demonstrated that both natural language queries and internal model activations encode predictive signals about factual accuracy and model uncertainty (e.g., Kadavath et al., 2022). As discussed in our introduction, inconsistencies in a model's answers to semantically related queries can also be leveraged to detect or mitigate hallucinations (Manakul et al., 2023; Xue et al., 2025; Agrawal et al., 2024).

Numerous other methods have proven effective in mitigating hallucinations; see, for example, the surveys by Ji et al. (2023) and Tian et al. (2024). In terms of evaluation, several comprehensive benchmarks and leaderboards have recently been introduced (e.g., Bang et al., 2025; Hong et al., 2024). However, relatively little work has examined barriers to their adoption. The 2025 AI Index report (Maslej et al., 2025), for instance, notes that hallucination benchmarks "have struggled to gain traction within the AI community."

Beyond binary expressions of certainty, more nuanced linguistic constructions have been proposed to communicate gradations of uncertainty (Mielke et al., 2022; Lin et al., 2022a; Damani et al., 2025). Additionally, the field of pragmatics—which investigates how meaning is shaped by context—has increasing relevance for understanding and improving how language models convey information (Ma et al., 2025).

3 Pretraining Errors

Pretraining produces a base language model $\hat{p}$ that approximates the distribution text drawn from its training distribution p. This is the classic "density estimation" problem in unsupervised learning, where a *density* is simply a probability distribution over data. In the case of language models, the distribution is over text or multimodal inputs if included.

The key challenge in proving that base models err is that many language models do not err. The degenerate model which always outputs IDK also avoids errors (assuming IDK is not an error). Similarly, assuming error-free training data, the trivial base model which regurgitates text from a random training example also does not err. However, these two language models fail at density estimation, the basic goal of statistical language modeling as defined below. Errors are also avoided by the optimal base model $\hat{p} = p$ which matches the training distribution, but this model would

require prohibitively large training data. Nonetheless, we show that well-trained base models should still generate certain types of errors.

Our analysis shows that generating valid outputs (i.e., avoiding errors) is harder than classifying output validity. This reduction enables us to apply the lens of computational learning theory, where errors are expected and understood, to error mechanisms in generative models. A language model is initially defined as a probability distribution over text and later *prompts* are incorporated (Section 3.2); both settings share the same intuition. Examples without prompts include birthday statements such as those of Fig. 1, while a prompted model might be queried for a specific individual's birthday.

Not merely autocomplete. Our analysis applies to general density estimation and not only "next-word predictors" even though many language models are trained using *self-supervised learning* to predict each word based on the previous words. It is tempting to attribute hallucinations to poorly chosen prefixes (e.g., "Adam Kalai was born on") for which the language model cannot provide valid completions. However, from a purely statistical perspective, ignoring computation, the autocomplete view[4] of language models is no more significant than the fact that any human speaker produces one word at a time. Our analysis suggests that errors arise from the very fact that the models are being fit to the underlying language distribution, though the specific architecture can introduce additional errors.

3.1 The reduction without prompts

Without prompts, a base model $\hat{p}$ is a probability distribution over a set $\mathcal{X}$. As discussed earlier, each *example* $x \in \mathcal{X}$ represents a "plausible" string, e.g., a document.[5] The examples $\mathcal{X} = \mathcal{E} \cup \mathcal{V}$ are partitioned into errors $\mathcal{E}$ and valid examples $\mathcal{V}$, for nonempty disjoint sets $\mathcal{E}, \mathcal{V}$. The error rate of base model $\hat{p}$ is denoted by,

$$\text{err} := \hat{p}(\mathcal{E}) = \Pr_{x \sim \hat{p}}[x \in \mathcal{E}]. \tag{1}$$

Training data are assumed to come from a noiseless *training distribution* $p(\mathcal{X})$, i.e., where $p(\mathcal{E}) = 0$. As discussed, with noisy training data and partly correct statements, one may expect *even higher error rates* than our lower bounds.

We now formalize the IIV binary-classification problem, introduced in the introduction. IIV is specified by the target function $f : \mathcal{X} \to \{-, +\}$ to be learned (membership in $\mathcal{V}$) and the distribution D over examples $\mathcal{X}$ (a 50/50 mix of samples from p and uniformly random errors):

$$D(x) := \begin{cases} p(x)/2 & \text{if } x \in \mathcal{V}, \\ 1/2|\mathcal{E}| & \text{if } x \in \mathcal{E}, \end{cases} \text{ and } f(x) := \begin{cases} + & \text{if } x \in \mathcal{V}, \\ - & \text{if } x \in \mathcal{E}. \end{cases}$$

Our analysis lower bounds the error rate $\text{err} = \hat{p}(\mathcal{E})$ in terms of IIV's aforementioned *misclassification rate* err_{iiv}:

$$\text{err}_{\text{iiv}} := \Pr_{x \sim D}\left[\hat{f}(x) \neq f(x)\right], \text{ where } \hat{f}(x) := \begin{cases} + & \text{if } \hat{p}(x) > 1/|\mathcal{E}|, \\ - & \text{if } \hat{p}(x) \leq 1/|\mathcal{E}|. \end{cases} \tag{2}$$

[4]Mathematically, any distribution p induces a distribution of completions $p(w_i w_{i+1} \ldots \mid w_1 w_2 \ldots w_{i-1})$ for every prefix of words $w_1 \ldots w_{i-1}$ in its support.

[5]We assume that $\mathcal{X}$ is finite for simplicity. See Section 5 for further discussion of errors and plausibility.